Dheeraj Reddy Jonnalagadda

+91-9381253827 |dheerajreddy_j@yahoo.in| linkedin.com/in/jonnalad | github.com/acentauri92

Summary

Experienced Embedded Software developer with 7 years of experience in designing and developing commercial products and prototypes.

Technical Skills

Languages: C, Python, Basic C++
Developer Tools: Git, CMake, Makefiles, gdb, Valgrind, Wireshark
Libraries: jansson, pandas, matplotlib
Unit Test Frameworks: Unity, Cpputest, CppUMock
RTOS: FreeRTOS, Micrium uC/OS III

Experience

Wipro Limited, Bengaluru, India
Senior Embedded Software Developer **04/2022 - Present**
Client: British Telecom

- Worked on developing firmware that runs on commercial internet routers. Much of my role has been to debug and optimize the current software performance on the router using multithreading (pthreads).
- Responsible for writing unit test cases for various firmware modules using Unity and CMock.

Client: Dell Technologies

- Developed networking software using sockets and TCP/IP that runs on the host operating system (Linux based) of the Dell PowerEdge servers to communicate with the add-on modules on the server that is based on libcurl.
- Validating software and making sure it is bug and leak free using tools like Valgrind, Checkmarx and purify.

Agnikul Cosmos, Chennai, India
Embedded Software Developer **01/2020 - 03/2022**

- Designing data acquisition software for STM32 microcontrollers using uC/OS III RTOS. The software collects data from multiple ADCs, monitors their range and packetizes them to be sent further using SPI to the main computer.
- Writing networking software for a distributed Ethernet network of about 10 nodes running a custom Linux kernel. Network communicates using UNIX sockets and TCP/IP and multithreading to implement the functionality. All code written in C.
- Building a custom Linux kernel for the NXP LS1028 processor for use in real-time environments using Buildroot. Key goal here was to build the kernel with the Preempt RT patch and minimize the image size.
- Network traffic simulation analysis using Omnet++. The traffic captured was analyzed and plotted using pandas and matplotlib.

Vanderbilt University Nashville, USA
Staff Research Engineer **07/2018 - 10/2019**

- Develop software to read and analyze MRI imaging data sent by the university hospital and display premade graphics on the university website. All software is developed in C and python.
- Developing automation software to monitor incoming MRI images Scripts written in python. This eliminates the manual process of constantly checking the website for new data.

Viaan Electronics (Rebranded as Viaanix) Kansas, USA
Embedded Software Developer **06/2017 - 05/2018**

- Planned, built, and developed a fully functional home security system based on the AM335x processor and the STM32Fxx series MCUs. Firmware to interface with Infrared sensors and cameras is written in C.
- Develop LoRa based wireless communication modules to track the movement of vehicles. The STM32F4 based system tracks movement and sends the information wirelessly using the SX1272 LoRa module back to the main hub. All functionality implemented using C and FreeRTOS. This product is now commercially used by companies across Wichita, Kansas.
- Collaborate with the hardware team to develop a graphical user interface (built using python and Tkinter) that would automatically perform tests on all the boards developed at the company. The software interacts with the fixtures using UART.

Education

Indiana University-Purdue University, Indianapolis, USA **01/2015 - 12/2016**
Master of Science (M.S), Electrical and Computer Engineering. GPA 3.0

SRM University, Chennai, India **07/2010 - 05/2014**
Bachelor of Technology (B. Tech), Electronics and Communication Engineering. GPA 7.8

Additional Interests

Hacking internet routers and switches

- Making hardware and software changes to get access to internet routers and customize them for projects.
- Build custom Linux kernels for the router.
- Built firmware images using OpenWRT and DDWRT. All my projects related to this are documented on Hackster.

Exploring the world around us through a phone

- Smartphones these days have lots of sensors in them and you can access all their data with the help of an app from Google. I ran quite a few experiments ranging from cabin pressure measurements in an aircraft to heartbeat variations.
- You can check out all the little experiments that I have done with the sensor data on my blog medium.